

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12397404

Kind Code

B2

Date of Patent

August 26, 2025

Inventor(s)

McConeghy; Ian et al.

Operating mode detection system for rotary hammer

Abstract

A rotary hammer includes a housing and a mode selection dial supported in the housing. The mode selection dial includes a cam. The housing supports a motor providing a rotational output. The housing supports a linkage moveable between a first position and a second position in response to engagement by the cam. A loss-of-control detection system measures acceleration of the housing and disables the motor upon the acceleration exceeding an acceleration threshold. The housing supports an operating mode detection system including a magnet and a Hall-effect sensor. The magnet is coupled to the linkage and moveable therewith between a third position corresponding to the first position of the linkage and a fourth position corresponding to the second position of the linkage. The Hall-effect sensor provides an output signal indicative of the position of the magnet. The loss-of-control detection system is selectively disabled based upon the output signal.

Inventors: McConeghy; Ian (Mequon, WI), Kendall; Bryan C. (Lisbon, WI), Gibb; Jesse (Milwaukee, WI), Spraggon; Jeffery D. (Pewaukee, WI), Lombardo; Matt (Muskego, WI), Wiehe; Hunter (Milwaukee, WI), Stingl; Spencer D. (Wauwatosa, WI), Rehfeldt; Alan J. (Oak Creek, WI)

Applicant: MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

Family ID: 1000008781010

Assignee: Milwaukee Electric Tool Corporation (Brookfield, WI)

Appl. No.: 18/669728

Filed: May 21, 2024

Prior Publication Data

Document Identifier

Publication Date

US 20240399550 A1

Dec. 05, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: B25D16/00 (20060101)

U.S. Cl.:

CPC **B25D16/006** (20130101); B25D2216/0084 (20130101); B25D2250/141 (20130101);
B25D2250/195 (20130101); B25D2250/221 (20130101)

Field of Classification Search

CPC: B25D (16/006); B25D (17/043); B25D (2250/221); B25D (2216/0023); B25D
(2250/255); B25D (2250/261); B25D (2216/0084); B25D (2250/121)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6766868	12/2003	Frauhammer et al.	N/A	N/A
6868919	12/2004	Manschitz et al.	N/A	N/A
7320368	12/2007	Watanabe	N/A	N/A
7398834	12/2007	Jung et al.	N/A	N/A
7845426	12/2009	Jung et al.	N/A	N/A
9010455	12/2014	Jung et al.	N/A	N/A
9180585	12/2014	Jung et al.	N/A	N/A
9254562	12/2015	Furusawa et al.	N/A	N/A
10322502	12/2018	Wirnitzer et al.	N/A	N/A
10414033	12/2018	Ekstrom et al.	N/A	N/A
10589413	12/2019	Goble	N/A	N/A
10615670	12/2019	Madineni et al.	N/A	N/A
10786893	12/2019	Nitsch	N/A	B25D 17/043
11059156	12/2020	Abe et al.	N/A	N/A
11192232	12/2020	Goble	N/A	N/A
11285594	12/2021	Gumpert et al.	N/A	N/A
11787030	12/2022	Wirnitzer et al.	N/A	N/A
2006/0185865	12/2005	Jung et al.	N/A	N/A
2009/0183888	12/2008	Forster et al.	N/A	N/A
2010/0289484	12/2009	Quinn	N/A	N/A
2012/0255752	12/2011	Aoki	N/A	N/A
2016/0089757	12/2015	Wirnitzer et al.	N/A	N/A
2018/0015603	12/2017	Abe	N/A	H02P 3/08
2020/0235643	12/2019	Madineni et al.	N/A	N/A
2021/0039231	12/2020	Araki et al.	N/A	N/A
2021/0039240	12/2020	Iida	N/A	N/A
2021/0078146	12/2020	Araki et al.	N/A	N/A

2022/0001526	12/2021	Wang et al.	N/A	N/A
2022/0105616	12/2021	Wirnitzer et al.	N/A	N/A
2022/0176534	12/2021	Billings et al.	N/A	N/A
2022/0219309	12/2021	Billings et al.	N/A	N/A
2022/0297279	12/2021	Abbott et al.	N/A	N/A
2022/0395971	12/2021	Iida et al.	N/A	N/A
2022/0395972	12/2021	Iida	N/A	B25D 16/006

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105880685	12/2015	CN	N/A
108453675	12/2017	CN	N/A
208289833	12/2017	CN	N/A
110524493	12/2018	CN	N/A
212445065	12/2020	CN	N/A
114074312	12/2021	CN	N/A
19720947	12/1996	DE	N/A
102019200522	12/2019	DE	N/A
2018020839	12/2017	WO	N/A
2022247813	12/2021	WO	N/A

Primary Examiner: Chukwurah; Nathaniel C

Attorney, Agent or Firm: Quarles & Brady LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/582,991 filed on Sep. 15, 2023 and U.S. Provisional Patent Application No. 63/505,023 filed on May 30, 2023, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present disclosure relates to rotary hammers, and more particularly to an operating mode detection system of the rotary hammer.

SUMMARY OF THE INVENTION

(2) The disclosure provides a rotary hammer configured to produce concurrent rotational and axial motion of a tool bit having a loss of control detection system and an operating mode detection system. The loss of control detection system is deactivated based on the operating mode of the tool, as detected by the operating mode detection system.

(3) The disclosure provides, in another aspect, a rotary hammer operable in a first mode in which only a hammering operation to reciprocate a tool bit along a drive axis is performed, and a second mode in which the tool bit is rotationally driven about the drive axis. The rotary hammer includes a housing, a mode selection dial, a motor, a linkage, a loss-of-control detection system, and an operating mode detection system. The mode selection dial is supported in the housing and is moveable among a plurality of positions indicative of the first and second modes. The mode selection dial includes a cam. The motor is supported in the housing and provides a rotational output. The linkage is supported in the housing and is moveable between first and second positions in response to engagement by the cam. The loss-of-control detection system is configured to measure acceleration of the housing and disable the motor upon the acceleration exceeding an

acceleration threshold. The operating mode detection system is supported in the housing and includes a magnet and a Hall-effect sensor. The magnet is coupled to the linkage and is moveable therewith between a third position corresponding to the first position of the linkage, and a fourth position, corresponding to the second position of the linkage. The Hall-effect sensor is coupled to the housing and provides an output signal that is indicative of the position of the magnet. The loss-of-control detection system is selectively disabled based on the output signal.

(4) The disclosure provides, in another aspect, a rotary hammer including a housing, a sensor coupled to the housing, a mode selection dial, a motor, a transmission, a linkage supported in the housing, a controller, and an operating mode detection system. The sensor is configured to provide a parameter signal indicative of a measured parameter of the housing about a drive axis. The mode selection dial includes a cam and is supported in the housing and is moveable among a plurality of positions indicative of first and second modes of operation. The motor is supported in the housing and provides a rotational output. The transmission is supported in the housing and is configured to receive the rotational output from the motor and selectively provides a rotational transmission output to the tool bit. The linkage is moveable between first and second positions in response to engagement by the cam. The controller includes a loss-of-control detection system configured to receive the parameter signal and compare the parameter signal to a parameter threshold. The controller is operable to disable the motor upon the parameter signal exceeding the parameter threshold. The operating mode detection system includes a magnet that is coupled to and moveable with the linkage, and a Hall-effect sensor coupled to the housing and provides an output signal indicative of the position of the magnet. The loss-of-control detection system is disabled upon disabling of the rotational transmission output to the tool bit.

(5) The disclosure provides, in another aspect, a rotary hammer including a housing, a mode selection dial supported in the housing, a motor, an impact mechanism, a linkage supported in the housing, a controller, and an operating mode detection system including a magnet and a Hall-effect sensor. The mode selection dial is rotatable among a plurality of positions indicative of the operating modes, including a first, hammering operation, and a second mode in which only a drilling operation is performed. The mode selection dial includes a cam. The motor is supported in the housing and provides a rotational output received by the impact mechanism which transfers the rotational output to successive reciprocations thereby performing the first mode. The impact mechanism is selectively activatable by the mode selection dial. The linkage is moveable between first and second positions based on engagement by the cam. The controller includes a loss-of-control detection system configured to disable the motor upon acceleration of the housing exceeding an acceleration threshold. The magnet is coupled to the linkage and is moveable therewith. The Hall-effect sensor is coupled to the housing and provides an output signal to the controller indicative of the position of the magnet. The Hall-effect sensor is closer to the magnet when the linkage is in the second position than when the linkage is in the first position.

(6) Other features and aspects of the subject matter will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a first embodiment of a rotary hammer according to the present disclosure.

(2) FIG. 2 is a section view of a portion of the rotary hammer of FIG. 1.

(3) FIG. 3 is a section view of a portion of the rotary hammer of FIG. 1, including the transmission.

(4) FIG. 4 is a section view of a portion of the rotary hammer of FIG. 1, including the transmission.

(5) FIG. 5 is a section view of a portion of the rotary hammer of FIG. 1, including the transmission.

(6) FIG. 6 is a section view of a portion of the rotary hammer of FIG. 1, including the transmission.

- (7) FIG. 7 is a section view of a portion of the rotary hammer of FIG. 1, including the operating mode detection system.
- (8) FIG. 8 is a section view of a portion of the rotary hammer of FIG. 1, including the operating mode detection system.
- (9) FIG. 9 is a perspective view of the operating mode detection system of the rotary hammer of FIG. 1.
- (10) FIG. 10 is a perspective view of a section of another embodiment of a rotary hammer.
- (11) FIG. 11 is top view of the mode selection dial of the rotary hammer of FIG. 10.
- (12) FIG. 12 is a section view of the rotary hammer of FIG. 10.
- (13) FIG. 13 is a perspective view of the rotary hammer of FIG. 10, including the operating mode detection system and mode selection dial.
- (14) FIG. 14 is a perspective view of the rotary hammer of FIG. 10, including the operating mode detection system and mode selection dial.
- (15) FIG. 15 is a section view of the rotary hammer of FIG. 10, including the operating mode detection system.
- (16) FIG. 16 is a section view of the rotary hammer of FIG. 10, including the operating mode detection system.
- (17) FIG. 17 is a section view of the rotary hammer of FIG. 10, including the operating mode detection system.
- (18) Before any embodiments of the subject matter are explained in detail, it is to be understood that the subject matter is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The subject matter is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

- (19) With reference to FIGS. 1 and 2, a first embodiment of a tool (e.g., a rotary hammer 10) that is operable to convert a rotational motion of a motor 14 into concurrent rotational and axial motion of a tool bit 18 along a drive axis A1 is illustrated. The illustrated rotary hammer 10 is operable in four modes: (1) a hammer-only mode in which only a hammering operation, that is, the tool bit 18 is only reciprocated along the drive axis A1, is performed by the rotary hammer 10; (2) a drill-only mode in which only a drilling operation, rotation of the tool bit 18 about the drive axis A1, is performed by the rotary hammer 10; (3) a hammer-drill mode, in which the hammering operation and the drill operation are performed concurrently by the rotary hammer 10; and (4) a chisel-adjustment, or chisel-rotate mode, in which the tool bit 18 is rotatable about the drive axis A1 by the user but rotational output of the motor 14 is not transmitted to the tool bit 18 and only the hammering operation is performed. The tool bit 18 is illustrated as a drill bit, however other types of bits (e.g., a chisel bit) can be substituted in place of the drill bit, depending on the application in which the user is operating the rotary hammer 10.
- (20) The rotary hammer 10 has a housing 22 with sides (left side 26 shown, right side not shown, but similar to left side), a top 34 joining the sides 26, and a front 38 from which tool bit 18 extends. The rear 42 of the housing 22 includes a handle 46 from which a trigger 50, which activates the rotary hammer 10, extends. A power source 54 (e.g., a battery pack, such as a rechargeable battery pack having a voltage capacity of 12V, 18V, or other voltage capacity) is coupled to the housing 22 adjacent the bottom 58 of the housing 22.
- (21) The housing 22 includes a lighting ring 62 (shown in FIGS. 7 and 8) disposed at the front 38 of the housing 22. The lighting ring 62 includes a plurality of light emitting diodes 66 (LEDs, e.g., three LEDs) each supported on a printed circuit board (LED PCB 70). The lighting ring 62 defines a channel 74 that runs along an extension portion 78 of the lighting ring 62 that extends from an annular portion 82 in which the LEDs 66 are located. Power wires 84 for the LEDs 66 are routed

within the channel **74** and interconnect the LED PCBs **70** and a controller **90**. The LEDs **66** are directed toward the tool bit **18** to illuminate the tool bit **18** and workspace surrounding the tool bit **18**.

(22) With reference to FIG. 2, the controller **90** is supported in the housing **22** in a direction rearward (i.e., toward the handle **46**, toward the left side of the page, as illustrated in FIG. 2) of the motor **14**. The controller **90** is configured to control operation of the rotary hammer **10**. The controller **90** includes a loss-of-control detection system **94** ("LOCDS," illustrated schematically) having a sensor **98** that measures a parameter of the housing **22** (e.g., the acceleration, angular acceleration, angular velocity, position of the housing **22**, etc.) and outputs a parameter signal. The controller **90** performs a protective operation (e.g., disables operation of the motor **14**, reduces power supplied to the motor **14**, etc.) when the measured parameter exceeds a parameter threshold for a set amount of time. The amount of time may vary depending on many factors (e.g., type of tool connected, selected operation mode, etc.).

(23) In that regard, while the rotary hammer **10** is operating, the motor **14** (e.g., a brushless DC motor) provides a rotational output that is transmitted to tool bit **18** and the controller **90** samples the parameter signal (e.g., the angular acceleration in each of three orthogonal principal axes) of the sensor **98** to determine the measured parameter (e.g., angular acceleration about the drive axis **A1** measured in radians per second-squared) of the housing **22**. In other embodiments, the controller **90** may integrate the output or perform other operations to determine a measured parameter (e.g., angular velocity of the tool **1**, measured in radians per second; position of the rotary hammer **10** relative to an initial position), and perform a protective operation when the measured parameter (e.g., angular velocity or the position of the rotary hammer **10**) has exceeded a tool threshold (e.g., threshold velocity, or a change in position that has exceeded a threshold). In some embodiments, the controller **90** samples the rotational speed of the housing **22** every ten milliseconds, but in other embodiments this sampling frequency can be higher or lower. For example, in some embodiments, the controller **90** samples the rotational speed of the housing **22** every millisecond.

(24) In some embodiments, the controller **90** increments or decrements a counter upon comparison of the measured parameter to the tool threshold and determination that the measured parameter has exceeded the tool threshold, and subsequently compares the counter to a counter threshold, and if the counter has exceeded the counter threshold, determines that a loss-of-control event has occurred, and performs a protective operation on the rotary hammer **10** when the counter has exceeded the counter threshold. In other embodiments, the controller **90** may instead decrease power supplied to the motor **14** to slow rotation of the motor **14**.

(25) The housing **22** supports the motor **14** which includes a rotor **102** coupled to a motor output shaft **106** that is supported within a stator **110** (e.g., via bearings **114**) and is rotatable about a motor rotation axis **A2**. A fan **118** is coupled to the motor output shaft **106** at the first end **122** and a motor pinion **126** is coupled to the motor output shaft **106** at the second end **130**. The motor pinion **126** engages a transmission input gear **134** (e.g., a bevel gear) to transfer rotation of the motor output shaft **106** to a transmission **138**.

(26) The transmission **138** includes an intermediate shaft **142** on which the transmission input gear **134**, a transmission pinion **146**, a rotational-output gear **150**, and first and second coupling sleeves **154**, **158** are supported. The transmission pinion **146** is coupled to the intermediate shaft **142** for rotation with the intermediate shaft **142** about a transmission rotation axis **A3** and the rotational-output gear **150** is supported on the intermediate shaft **142** (e.g., by bearings) such that the rotational-output gear **150** is rotatable relative to, that is, independent from, the intermediate shaft **142**. The first coupling sleeve **154** is supported on the transmission pinion **146** and is slidable to engage the rotational-output gear **150**, coupling the rotational-output gear **150** for rotation with the intermediate shaft **142** and transmission pinion **146**. The second coupling sleeve **158** is also supported on the transmission pinion **146** and is slidable relative to the transmission pinion **146** to engage an impact mechanism **162** supported on the intermediate shaft **142**.

(27) The rotational-output gear **150** engages a gear portion **166** of the spindle **170** to transfer rotation of the intermediate shaft **142** to the spindle **170**, and thereby, the tool bit **18** coupled to the spindle **170** (e.g., via a chuck, quick-change collet, or other securing structure configured to receive and couple a tool bit **18** to the rotary hammer **10**). Upon engagement of the first coupling sleeve **154** with the rotational-output gear **150**, the rotational output of the motor **14** is transferred from the motor output shaft **106** to the intermediate shaft **142** via the transmission input gear **134**, from the intermediate shaft **142** to the rotational-output gear **150** through the coupling engagement of the first coupling sleeve **154** with the transmission pinion **146** and rotational-output gear **150**, and from the rotational-output gear **150** to the spindle **170** and the tool bit **18**. Disengagement of the first coupling sleeve **154** from the rotational-output gear **150** disables the transfer of rotational output from the motor **14** to the tool bit **18**.

(28) The impact mechanism **162** (e.g., a wobble drive system including a wobble bearing **174**) is supported on the intermediate shaft **142** between the transmission input gear **134** and the transmission pinion **146**, with the rotational-output gear **150** rotatably supported on the intermediate shaft **142** in a position furthest away from the transmission input gear **134**. The impact mechanism **162** is also supported on the intermediate shaft **142** (e.g., with bearings) such that it is rotatable independent of the intermediate shaft **142**. The second coupling sleeve **158** slides on the transmission pinion **146** to engage the wobble bearing **174**. The impact mechanism **162** further includes a cylinder **178** supported in the spindle **170** and coupled to the wobble bearing **174**. The cylinder **178** is reciprocated by the wobble bearing **174**, and a striker **182** supported in the cylinder **178** is reciprocated by an air cushion within the cylinder **178** between the cylinder **178** and the striker **182**. The striker **182** impacts an anvil **186** supported in the spindle **170** and imparts axial impacts thereon, which are transmitted to the tool bit **18** as a hammering impact.

(29) A mode selection dial **86** is supported on one side **26** of the housing **22** and is rotatable among a plurality of positions (P1-P4) that indicate the mode in which the rotary hammer **10** is being operated. With reference to FIGS. 3-6, the mode selection dial **86** includes first and second arms **190**, **194** extending from the mode selection dial **86** and a cam **198** at least partially defined by surfaces of the mode selection dial **86** and the arms **190**, **194**. The arms **190**, **194** engage the first and second coupling sleeves **154**, **158** to translate the first and second coupling sleeves **154**, **158** along the transmission pinion **146**.

(30) A linkage **202** is supported in the housing **22** and is slidable parallel to the drive axis A1 between a first and second position P5, P6. A biasing member **206** (e.g., a compression spring) biases the linkage **202** to the second position P6 (FIG. 3). The cam **198** selectively engages the linkage **202** (e.g., by rotation of mode selection dial **86**) to overcome the bias of the biasing member **206** and translate the linkage **202** to the first position P5 (FIG. 4). With reference to FIG. 9, the linkage **202** includes an engagement portion **210** having teeth **214** corresponding to the pitch of the rotational-output gear **150**. Returning to FIG. 3, the linkage **202** engages the rotational-output gear **150** when the linkage **202** is in the second position P6, preventing rotation of the rotational-output gear **150**, and thereby, rotation of the spindle **170**.

(31) With reference to FIG. 3, when the mode selection dial **86** is the position P1 corresponding to the first, hammer-only mode in which only a hammering operation is performed by the rotary hammer **10**, the first arm **190** engages the first coupling sleeve **154** to slide the first coupling sleeve **154** along the transmission pinion **146** against the bias of a spring **218** that engages the first coupling sleeve **154** to slide the first coupling sleeve **154** out of engagement with the rotational-output gear **150**. By disengaging the transmission pinion **146** and rotational-output gear **150**, rotation is not transferred from the motor **14** to the spindle **170**, thereby disabling rotation of the tool bit **18**. The spring **218** biases the second coupling sleeve **158** along the transmission pinion **146** into engagement with the wobble bearing **174**, allowing the rotational output of the motor **14** to be transferred from the intermediate shaft **142** and transmission pinion **146** to the wobble bearing **174**, thereby enabling axial impact to be imparted to the tool bit **18**. In the position P1 corresponding to

the hammer-only mode, the cam **198** does not engage the linkage **202** and the linkage **202** is biased from the first position **P5** to the second position **P6** by the biasing member **206**. The engagement portion **210** of the linkage **202** is brought into contact with the rotational-output gear **150** and prevents rotation of the rotational-output gear **150**, and thus the spindle **170**.

(32) With reference to FIG. **4**, when the mode selection dial **86** is in the position **P2** corresponding to the second, drill-only mode in which only a drilling operation, or rotation of the tool bit **18** about the drive axis **A1**, is performed by the rotary hammer **10**. The second arm **194** of the mode selection dial **86** engages the second coupling sleeve **158** to slide the second coupling sleeve **158** along the transmission pinion **146** toward the front **38** of the rotary hammer **10** and out of engagement with the wobble bearing **174**. The spring **218** biases the first coupling sleeve **154** along the transmission pinion **146** into engagement with the rotational-output gear **150**, to enable the transfer of rotational output from the motor **14** to the rotational-output gear **150** via the transmission pinion **146**, and thereby, to the spindle **170** and tool bit **18**. The cam **198** of the mode selection dial **86** engages the linkage **202** to push the linkage **202** against the force of the biasing member **206** out of engagement with the rotational-output gear **150**, allowing rotation of the rotational-output gear **150**.

(33) With reference to FIG. **5**, when the mode selection dial **86** is the position **P3** corresponding to the third, hammer-drill mode, in which the hammering operation and the drill operation are performed concurrently, the arms **190**, **194** of the mode selection dial **86** do not engage the first and coupling sleeves **154**, **158**. The first coupling sleeve **154** engages the transmission pinion **146** and the rotational-output gear **150** to transfer rotation of the motor **14** to the rotational-output gear **150**, the spindle **170**, and the tool bit **18**. The second coupling sleeve **158** engages the wobble bearing **174** and transmission pinion **146** to transfer the rotational output of the motor **14** to the wobble bearing **174** which imparts an axial impact on the tool bit **18** via the impact mechanism **162**. The cam **198** biases the linkage **202** out of engagement with the rotational-output gear **150**.

(34) With reference to FIG. **6**, when the mode selection dial **86** is the position **P4** corresponding to the fourth, chisel-adjustment, or chisel-rotate mode, the first arm **190** of the mode selection dial **86** biases the first coupling sleeve **154** out of engagement with the rotational-output gear **150** thereby disabling the transfer of rotational output from the motor **14** to the rotational-output gear **150**, spindle **170**, and tool bit **18**. The cam **198** is engaged with the linkage **202** to bias the linkage **202** out of engagement with the rotational-output gear **150**, allowing rotation of the rotational-output gear **150** relative to the intermediate shaft **142**, without consequent transfer of rotation to the transmission pinion **146**. A user is able to rotate the spindle **170** and thus the tool bit **18** to position the tool bit **18** to perform an operation.

(35) With reference to FIGS. **7-9**, an operating mode detection system **222** is supported in the housing **22**. The operating mode detection system **222** includes a magnet **226** and a Hall-effect sensor **230** on to a Hall-effect PCB **232** that senses the magnetic field of the magnet **226** and outputs an output signal that is indicative of the position of the magnet **226** relative to the Hall-effect sensor **230**. In the present embodiment, the sensor is a pulse-width-modulated sensor that outputs a digital signal (i.e., a high or low value) with a duty cycle (i.e., pulse width) that is dependent on the strength of the magnetic field. In other embodiments, the sensor may be an analog sensor that outputs a signal that is linearly dependent on the strength of the magnetic field measured. In still other embodiments, the sensor may instead be a digital sensor that outputs either a high or low value, depending on the strength of the magnetic field being above, below, or within a sensor threshold.

(36) The magnet **226** is coupled to the linkage **202** and is slidable between a third position **P7** that corresponds to the first position **P5** of the linkage **202** (FIG. **7**), and a fourth position **P8** that corresponds to the second position **P6** of the linkage **202** (FIG. **8**). A magnet holder **234** (FIG. **9**) is coupled to the linkage **202** (e.g., in a snap fit), to maintain the position of the magnet **226** in coupled relationship with the linkage **202**. In other embodiments, the magnet **226** may be coupled

to the magnet holder **234** or the linkage **202** in another manner. When the linkage **202** is in the first position **P5** and the magnet **226** is in the third position **P7**, the magnet **226** is further from the Hall-effect sensor **230** than when the linkage **202** is in the second position **P6** and the magnet **226** is in the fourth position **P8**. The strength of the magnetic field, and therefore, the output signal, are stronger when the linkage **202** is in the second position **P6** and the magnet **226** is in the fourth position **P8** than when the linkage **202** is in the first position **P5** and the magnet **226** is in the third position **P7**.

(37) The Hall-effect PCB **232** is disposed in a receptacle **233** in the front **38** of the housing **22** and is electrically coupled to, and provides an output signal to the controller **90** via wires **238** that are disposed in the channel **74** of the lighting ring **62** alongside the power wires **84** for the LEDs **66**. In another embodiment, the receptacle **233** is defined in the extension portion **78** of the lighting ring **62**.

(38) The controller **90** disables the LOCDS **94** based on the output signal from the Hall-effect sensor **230**. In the present embodiment, the LOCDS **94** is operative as a default condition. That is, the LOCDS **94** is enabled unless the controller **90** has disabled the LOCDS **94**. The controller **90** disables the LOCDS **94** when the output signal from the Hall-effect sensor **230** is within a sensor threshold (i.e., within a range of values corresponding to the sensor threshold). In other embodiments, the controller **90** disables the LOCDS **94** when the output signal from the Hall-effect sensor **230** is below the sensor threshold. In other embodiments, the controller **90** disables the LOCDS **94** when the output signal from the Hall-effect sensor **230** is above the sensor threshold. As the linkage **202** is translated in a direction parallel to the drive axis **A1** from the first position **P5** to the second position **P6** and the magnet **226** is translated from the third position **P7** to the fourth position **P8**, the strength of the magnet field increases and the output signal of the Hall-effect sensor **230** reflects the increase in the magnetic field. The controller **90** disables the LOCDS **94** when the output signal is within the sensor threshold. As the linkage **202** is translated from the second position **P6** to the first position **P5**, and the magnet **226** from the fourth position **P8** to the third position **P7**, the magnetic field measured by the Hall-effect sensor **230** becomes weaker, or decreases to zero magnetic field, the output signal of the Hall-effect sensor **230** reflects the decrease, and the LOCDS **94** defaults to the enabled state. Stated another way, the controller **90** disables the LOCDS **94** when the rotary hammer **10** is operated in the hammer-only and chisel adjustment modes, that is, when the rotational transmission output to the tool bit **18** is disabled. The risk of the operator losing control of the rotary hammer **10** is reduced when rotation is not transmitted to the spindle **170**. It will be appreciated that, by disabling the LOCDS **94**, nuisance shut-offs (e.g., protective operations in which the motor is disabled while the user maintains, that is, has not lost, control of the rotary hammer) can be eliminated.

(39) With reference to FIGS. **10-17**, a second embodiment of a rotary hammer **10'** (e.g., a rotary hammer) that is operable to convert a rotational motion of a motor **14'** into concurrent rotational and axial motion of a tool bit **18'** along a drive axis **A1** is illustrated, similar to the rotary hammer **10** above. Numbering of features differing from those above will include a prime (') designation, with like features being represented with like reference numerals.

(40) With reference to FIGS. **10-12**, the illustrated rotary hammer **10'** is operable in three modes: (1) a hammer-only mode in which only a hammering operation, that is, the tool bit **18'** is only reciprocated along the drive axis **A1**, is performed by the rotary hammer **10'**; (2') a hammer-drill mode, in which the hammering operation and the drill operation are performed concurrently by the rotary hammer **10'**; and (3') a chisel-adjustment, or chisel rotate mode, in which the tool bit **18'** is rotatable about the drive axis **A1** by the user but rotational output of the motor **14'** is not translated to the tool bit **18'** and only the hammering operation is performed.

(41) Instead of the mode selection dial **86'** rotatably supported in a side **26'** of the housing **22'**, the mode selection dial **86'** is supported at the top **34'** of the housing **22'**. The mode selection dial **86'** is rotatable among four positions (**P1'-P4'**) indicative of the three operating modes.

(42) The controller **90'** is supported in the housing **22'** in a direction below (i.e., toward the bottom of the page, as illustrated in FIG. **10**) the motor **14'**, away from the mode selection dial **86'**. The controller **90'** is configured to control operation of the rotary hammer **10'** in substantially the same manner as the first embodiment. In that regard, the controller **90'** includes a LOCDS **94** (illustrated schematically) and performs a protective operation, for instance, disabling the motor **14'**, reducing power to the motor **14'**, etc., when the measured parameter of a sensor **98** exceeds a parameter threshold.

(43) The motor **14'** includes a rotor **102'** coupled to a motor output shaft **106'** that is rotatably supported within a stator **110'** (e.g., via bearings **114**). A fan **118'** is coupled to the motor output shaft **106'** at the second end **130** and a motor pinion **126'** is also coupled to the motor output shaft **106'** at the second end **130**. The motor pinion **126'** engages a transmission input gear **134'** to transfer rotation of the motor output shaft **106'** to a transmission **138'** and to an impact input gear **242** to transfer rotation to the impact mechanism **162'**.

(44) The transmission **138'** includes an intermediate shaft **142'** on which the transmission input gear **134'** is supported. The intermediate shaft **142'** is rotatably supported in the housing **22'** (e.g., by bearings **144'**) and includes a rotational-output gear portion **150'** that engages a gear portion **166'** of the spindle **170'** to transfer rotational output of the motor **14'** to the spindle **170'** and tool bit **18'** coupled thereto. The intermediate shaft **142'** defines a transmission rotational axis **A3** that is substantially parallel to the rotational axis **A2** of the motor **14'** and perpendicular to the drive axis **A1**.

(45) The impact mechanism **162'** includes the impact input gear **242** coupled to a crankshaft **246** at the first end **250** of the crankshaft **246**. The crankshaft **246** includes an eccentric pin **254** at the second end **258** that is coupled to a connecting rod **262**. The connecting rod **262** is coupled to a piston **266** slidably supported in the spindle **170'**. Rotational output of the motor **14'** is transferred to the impact mechanism **162'** via the impact input gear **242**, which in turn rotates the crankshaft **246** about a crankshaft axis **A4**. Rotation of the crankshaft **246** about the crankshaft axis **A4** is transferred to the connecting rod **262** by the eccentric pin **254** which translates rotation of the motor **14'** and crankshaft **246** to axial movement of the connecting rod **262**, and therefore the piston **266**, along the drive axis **A1**. The piston **266** transfers axial movement to a striker **270** supported in the spindle **170'** via an air cushion between the piston **266** and striker **270**, which is reciprocated along the drive axis **A1** with the piston **266**. The striker **270** contacts an anvil **186'** to impart successive impacts on the anvil **186'**, and consequently, the tool bit **18'**.

(46) With reference to FIGS. **13-14**, the mode selection dial **86'** includes a cam **198'** that is eccentrically oriented. A linkage **202'** having a central opening **274** is slidably supported in the housing **22'** and the cam **198'** is positioned in the central opening **274**. The eccentric positioning of the cam **198'** in the central opening **274** drives the linkage **202'** to slide in a side-to-side direction along a linkage movement axis **A5** that is perpendicular to the drive axis **A1** between a first position **P5'**, illustrated in the figures as a furthest rightward position, and a second position **P6'**, illustrated in the figures as a furthest leftward position.

(47) With reference to FIGS. **13** and **15**, when the mode selection dial **86'** is the position **P1** corresponding to the first, hammer-only mode in which only a hammering operation is performed by the rotary hammer **10'**, the linkage **202'** is positioned in the first, rightward position **P5'** by the cam **198'** positioned in the central opening **274**. In the hammer-only mode, rotational output to the tool bit **18'** is disabled and only axial impacts are imparted to the tool bit **18'**.

(48) With reference to FIGS. **14** and **16**, when the mode selection dial **86'** is the position **P2'** corresponding to the second, hammer-drill mode, in which the hammering operation and the rotational drilling operation are performed concurrently, the linkage **202'** is in the second, furthest leftward position **P6'** based on the position of the cam **198'** in the central opening **274**. In the hammer-drill mode, rotational output to the tool bit **18'** is enabled, along with axial impact applied to the tool bit **18'**.

(49) With reference to FIGS. 11 and 17, when the mode selection dial 86' is in the positions P3', P4' corresponding to the third, chisel-adjustment mode, the linkage 202' is positioned between the first and second positions P5', P6' based on the position of the cam 198' in the central opening 274. In the chisel adjustment mode, rotational output from the motor 14' to the tool bit 18' is disabled, but the tool bit 18' can be rotatably positioned.

(50) With reference to FIGS. 13-17, an operating mode detection system 222' is supported in the housing 22'. The operating mode detection system 222' includes a magnet 226' and a Hall-effect sensor 230', coupled to a Hall-effect PCB 232', that operate in substantially the same manner as described for the previous embodiment.

(51) The magnet 226' is coupled to the linkage 202' and is slidable between a third position P7' that corresponds to the first position P5' of the linkage 202' (FIGS. 13, 15), and a fourth position P8' that corresponds to the second position P6' of the linkage 202' (FIGS. 14, 16). A magnet holder 234' is coupled to the linkage 202' (e.g., in a snap fit), to maintain the position of the magnet 226' in coupled relationship with the linkage 202'.

(52) The Hall-effect sensor 230' is coupled to housing 22' the adjacent the linkage 202'. The Hall-effect sensor 230' provides an output signal to the controller 90'. When the linkage 202' is in the first position P5' and the magnet 226' is in the third position P7', the magnet 226' is further from the Hall-effect sensor 230' than when the linkage 202' is in the second position P6' and the magnet 226' is in the fourth position P8'. The strength of the magnetic field, and therefore, the output signal, are stronger when the linkage 202' is in the second position P6' and the magnet 226' is in the fourth position P8' than when the linkage 202' is in the first position P5' and the magnet 226' is in the third position P7'.

(53) The controller 90' enables the LOCDS 94' based on the output signal from the Hall-effect sensor 230'. In the present embodiment, the LOCDS 94' is disabled as a default condition. That is, the LOCDS 94' is disabled unless the controller 90' has enabled the LOCDS 94'. Stated another way, the controller 90' enables the LOCDS 94' when the rotary hammer 10 is operated in the rotary hammer mode, that is, when the rotational transmission output to the tool bit 18 is enabled; the LOCDS 94' is disabled when transmission rotational output to the tool bit 18 is disabled. The controller 90' enables the LOCDS 94' when the output signal from the Hall-effect sensor 230' is within a sensor threshold. As the linkage 202' is translated in along the linkage movement axis A5 perpendicular to the drive axis A1 from the first position P5' to the second position P6' and the magnet 226' is translated from the third position P7' to the fourth position P8', the strength of the magnetic field increases and the output signal of the Hall-effect sensor 230' reflects the increase in the magnetic field. The controller 90' enables the LOCDS 94' when the output signal is within the sensor threshold. As the linkage 202' is translated from the second position P6' to the first position P5', and the magnet 226' from the fourth position P8' to the third position P7', the magnetic field measured by the Hall-effect sensor 230' becomes weaker, or decreases to zero magnetic field, and the output signal of the Hall-effect sensor 230' reflects the decrease, and the LOCDS 94' returns to the default disabled state.

(54) The controller 90, 90' of any of the previous embodiments may control other tool parameters based on the output signal from the Hall-effect sensor 230, 230'. In one embodiment, the rotational direction of the motor 14, 14' can be controlled by the controller 90, 90' based on the output signal received from the Hall-effect sensor 230, 230'. That is, when the Hall-effect sensor 230, 230' provides a signal indicative of the rotary hammer 10, 10' operating in the hammer-only mode, the controller 90, 90' enable motor rotation in only one direction. In another embodiment, the trigger 50 is a variable-speed trigger whereby the rotational speed of the motor 14, 14' is increased as the trigger 50 is depressed a greater distance, and the trigger mapping is modified by the controller 90, 90' based on the output signal of the Hall-effect sensor 230, 230'. For instance, when the controller 90, 90' determines the rotary hammer 10, 10' is operating in the hammer-only mode based on the output signal from the Hall-effect sensor 230, 230', the controller 90, 90' operates the motor 14, 14'

at a maximum rotational speed when the trigger **50** has been depressed a smaller amount than when the rotary hammer **10, 10'** is operating in another mode (e.g., drill only mode, rotary-hammer mode). Stated another way, the motor **14, 14'** is operated at the maximum rotational speed with a greater trigger displacement in one operating mode than in a different operating mode. As an example, when the rotary hammer **10, 10'** is operated in the hammer-only mode, the motor **14, 14'** may be operated at the maximum rotational speed when the trigger **50** has been depressed fifty percent of the maximum trigger displacement, and when the rotary hammer **10, 10'** is operated in other modes, the motor **14, 14'** may be operated at the maximum rotational speed when the trigger **50** has been depressed seventy-five percent of the maximum trigger displacement. In other embodiments, different trigger mapping may be used, for instance, different trigger displacement percentages, different operating modes, etc. In other embodiments, the controller **90, 90'** may operate other parameters of the rotary hammer **10, 10'** in a different manner depending on the output signal from the Hall-effect sensor **230, 230'**.

(55) Although the subject matter has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the subject matter as described.

(56) Various features of the subject matter are set forth in the following claims.

Claims

1. A rotary hammer operable in a first mode in which only a hammering operation to reciprocate a tool bit along a drive axis is performed and a second mode in which the tool bit is rotationally driven about the drive axis, the rotary hammer comprising: a housing; a mode selection dial supported in the housing and moveable among a plurality of positions indicative of the first mode and the second mode, the mode selection dial including a cam; a motor supported in the housing, the motor providing a rotational output; a linkage supported in the housing and moveable between a first position and a second position in response to engagement by the cam; a loss-of-control detection system configured to measure acceleration of the housing and disable the motor upon the acceleration exceeding an acceleration threshold; and an operating mode detection system supported in the housing, the operating mode detection system including: a magnet coupled to the linkage and moveable therewith between a third position corresponding to the first position of the linkage and a fourth position corresponding to the second position of the linkage, and a Hall-effect sensor coupled to the housing, the Hall-effect sensor providing an output signal indicative of a position of the magnet, wherein the loss-of-control detection system is selectively disabled based upon the output signal.
2. The rotary hammer of claim 1, wherein the operating mode detection system further includes a magnet holder coupled to the linkage, the magnet holder securing the magnet to the linkage.
3. The rotary hammer of claim 2, wherein the magnet holder is coupled to the linkage via a snap fit.
4. The rotary hammer of claim 1, further comprising a controller configured to operate the loss-of-control detection system, wherein the controller is coupled to the Hall-effect sensor by a first wire, and wherein the housing defines a channel in which the first wire is at least partially disposed.
5. The rotary hammer of claim 4, further comprising a lighting ring disposed at a front side of the housing and electrically connected to the controller by a second wire, wherein the second wire is at least partially disposed within the channel with the first wire.
6. The rotary hammer of claim 1, wherein the mode selection dial is supported on a side of the housing.
7. The rotary hammer of claim 1, wherein the mode selection dial is supported on a top of the housing.
8. The rotary hammer of claim 1, wherein the linkage is movable in a direction parallel to the drive axis.

9. The rotary hammer of claim 1, further comprising: an intermediate shaft defining a rotational axis, the intermediate shaft supported in the housing such that the rotational axis is perpendicular to the motor rotational output and parallel to the drive axis, the intermediate shaft receiving the rotational output from the motor; a pinion coupled to the intermediate shaft and rotatable therewith; a rotational-output gear supported on the intermediate shaft and rotatable relative thereto, the rotational-output gear configured to provide a rotational output to the tool bit, the linkage engageable with the rotational-output gear in the second position; and a coupling sleeve slidably disposed on the pinion and selectively engageable with the rotational-output gear to transmit rotation from the pinion to the rotational-output gear, wherein the mode selection dial engages the coupling sleeve to slide the coupling sleeve along the pinion out of engagement with the rotational-output gear concurrently with the linkage moving to the second position thereby engaging the rotational-output gear and disabling rotational output to the tool bit.

10. The rotary hammer of claim 1, wherein the loss-of-control detection system is disabled when the output signal is within a sensor threshold.

11. The rotary hammer of claim 1, wherein the fourth position is closer to the Hall-effect sensor than the third position, and wherein the loss-of-control detection system is enabled when the magnet is closer to the third position.

12. A rotary hammer operable in a first mode in which only a hammering operation to reciprocate a tool bit along a drive axis is performed and a second mode in which the tool bit is rotationally driven about the drive axis, the rotary hammer comprising: a housing; a sensor coupled to the housing and configured to provide a parameter signal indicative of a measured parameter of the housing about the drive axis; a mode selection dial supported in the housing and movable among a plurality of positions indicative of the first mode and the second mode, the mode selection dial including a cam; a motor supported in the housing, the motor providing a rotational output; a transmission supported in the housing and configured to receive the rotational output from the motor, the transmission configured to selectively provide a rotational transmission output to the tool bit; a linkage supported in the housing and moveable between a first position and a second position in response to engagement by the cam; a controller including a loss-of-control detection system configured to receive the parameter signal and compare the parameter signal to a parameter threshold, the controller operable to disable the motor upon the parameter signal exceeding the parameter threshold; and an operating mode detection system including: a magnet coupled to the linkage and movable therewith, and a Hall-effect sensor coupled to the housing, the Hall-effect sensor providing an output signal indicative of a position of the magnet, wherein the loss-of-control detection system is disabled upon disabling of the rotational transmission output to the tool bit.

13. The rotary hammer of claim 12, wherein the transmission includes: an intermediate shaft defining a rotational axis, the intermediate shaft supported in the housing such that the rotational axis is perpendicular to the motor rotational output and parallel to the drive axis, the intermediate shaft receiving the rotational output from the motor; a pinion coupled to the intermediate shaft and rotatable therewith; a rotational-output gear supported on the intermediate shaft and rotatable relative thereto, the rotational-output gear configured to provide a rotational output to the tool bit; and a coupling sleeve slidably disposed on the pinion and selectively engageable with the rotational-output gear to transmit rotation from the pinion to the rotational-output gear, wherein the linkage is slidable in a direction parallel to the rotational axis and engageable with the rotational-output gear in the second position, and wherein the mode selection dial engages the coupling sleeve to slide the coupling sleeve along the pinion out of engagement with the rotational-output gear concurrently with the linkage moving to the second position thereby engaging the rotational-output gear and disabling rotational output to the tool bit.

14. The rotary hammer of claim 13, wherein the linkage is movable in a direction perpendicular to the drive axis.

15. The rotary hammer of claim 12, wherein the loss-of-control detection system is disabled when

the output signal is within a sensor threshold.

16. The rotary hammer of claim 12, wherein the housing includes a lighting ring disposed at a front side of the housing, the lighting ring defining a channel therein extending away from the tool bit, the Hall-effect sensor coupled to the housing adjacent to the channel.

17. The rotary hammer of claim 12, wherein the operating mode detection system further includes a magnet holder coupled to the linkage, the magnet holder maintaining the magnet in coupled relation with the linkage.

18. A rotary hammer comprising: a housing; a mode selection dial supported in the housing and rotatable among a plurality of positions indicative of a plurality of modes including a first mode in which only a hammering operation to reciprocate a tool bit along a drive axis is performed and a second mode in which only a drilling operation to rotate the tool bit about the drive axis is performed, the mode selection dial including a cam; a motor supported in the housing and providing a rotational output; an impact mechanism supported in the housing and configured to receive the rotational output from the motor and transfer the rotational output to successive reciprocations thereby performing the first mode, the impact mechanism selectively activatable by the mode selection dial; a linkage supported in the housing and moveable between a first position and a second position based upon engagement by the cam; a controller including a loss-of-control detection system configured to disable the motor upon acceleration of the housing exceeding an acceleration threshold; and an operating mode detection system including: a magnet coupled to the linkage and moveable therewith, and a Hall-effect sensor coupled to the housing, the Hall-effect sensor providing an output signal to the controller indicative of a position of the magnet, wherein the Hall-effect sensor is closer to the magnet when the linkage is in the second position than when the linkage is in the first position.

19. The rotary hammer of claim 18, wherein the output signal is useable by the controller to control a rotational direction of the motor.

20. The rotary hammer of claim 19, wherein the output signal is useable by the controller to control rotational speed of the motor.
